

Ratio Monotonicity for a Killed von Mises Bridge:

exact bridge structure, certified negative results, a single-Bessel reduction, and a two-scale closure map for a surface expansion in two-dimensional lattice gauge theory

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Abstract

For $\beta > 0$ let $I_m = I_m(\beta)$ denote modified Bessel functions of the first kind, and set $a_m = I_m^2[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2]$, $b_m = mI_m^4$, $F_A(t) = \sum_{m \geq 1} a_m \sin(mt)$, $F_B(t) = \sum_{m \geq 1} b_m \sin(mt)$, and $E = F_A/(2F_B)$. The global ratio-monotonicity problem for the surface expansion of a two-dimensional $SU(2)$ lattice gauge observable is: (i) $F_B > 0$ on $(0, \pi)$, and (ii) $E' < 0$ on $(0, \pi)$, for every $\beta > 0$. We prove (i) twice. We prove a family of exact structural results: E is, *as an algebraic identity*, the mean of $\cos \psi$ under the midpoint law of a four-step killed von Mises bridge; the generating kernels reduce, via the Neumann addition theorem, to two-dimensional integrals of a *single* Bessel function with explicit trigonometric phase whose saddle deficit is an exact sum of two squares; and both E and E' admit exact α -direction cancellations at the saddle which force the closed second-order law $E = \cos(t/2)(1 - c(t)/\beta) + O(\beta^{-2})$ with $c(t) = (4 \cos^2(t/4) - 1)/(2 \cos(t/4) \cos(t/2))$. We prove (ii) for $0 < \beta \leq 3$ (an exact pair-identity skeleton plus one archived interval certificate), and prove three *certified negative results* (interval arithmetic, two independent implementations, nested enclosures): the time-marginals of the bridge one smoothing step from the moving endpoint are *not* stochastically monotone — killing every monotone full-path coupling — with the minimal counterexample at the two-step bridge midpoint, margin 1.3×10^{-2} . The mechanism is an exact limit identity with threshold $\beta |\cos t| = 3/2$. For the remaining range we give a two-scale closure map with all constants of our own manufacture, a candidate threshold $\beta_0 = 15$ whose truth map is verified at high precision, and the π -endpoint package: $A_\infty = 0$ and a telescoped identity for the cubic coefficient c_3 , with the verified prefactor law $c_3 \sim \beta^{3/2} e^{2\sqrt{2}\beta} / (24 \cdot 2^{1/4} \pi^{3/2})$ whose two-saddle factor 2 is an exact parity symmetry. Every claim is labelled exact / certified / verified; the machine-checked lemmas are Lean 4/Mathlib, machine-checked modulo classical Bessel inputs carried as named hypotheses.

1 Introduction

This paper consolidates the resolution status of a two-part positivity problem that arises as the last analytic gap in a surface (character) expansion for two-dimensional $SU(2)$ lattice gauge theory [5, 6, 7]. The problem is stated entirely in terms of modified Bessel functions and trigonometric series, and we treat it as such; no claim of physical consequence beyond the stated expansion is made, and importance is not inherited by thematic proximity.

Throughout, $\beta > 0$, $I_m = I_m(\beta)$, and

$$a_m = I_m^2[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2], \quad b_m = mI_m^4, \quad m \geq 1. \quad (1)$$

Both sequences decay super-exponentially, so

$$F_A(t) = \sum_{m \geq 1} a_m \sin(mt), \quad F_B(t) = \sum_{m \geq 1} b_m \sin(mt), \quad E(t) = \frac{F_A(t)}{2F_B(t)}, \quad (2)$$

are real-analytic on $\mathbb{R}$, and the Wronskian $W = 2(F'_A F_B - F_A F'_B)$ satisfies $W = 4F_B^2(E)'$ wherever $F_B \neq 0$ [7].

Conjecture 1 (global ratio monotonicity). *For every $\beta > 0$: (i) $F_B > 0$ on $(0, \pi)$; (ii) $E' < 0$ on $(0, \pi)$.*

Part (i) is proved below (twice). Part (ii) is proved for $0 < \beta \leq 3$ (Theorem 7; exact skeleton plus the archived certificate `certify_thmB.py`), is supported by an exact structural reduction and a verified truth map for $\beta \geq 15$, and is otherwise open; Section 8 states the precise two-scale closure map. The conjecture was born inside this programme (it is the global form of an ε -window positivity required by the surface expansion); no external pedigree is claimed.

Convention 2 (tricotomy). Every assertion below carries one of three labels. *Exact*: a proved identity or inequality. *Certified*: a numerical statement enclosed in interval arithmetic with outward rounding, nested re-computation at higher precision, and (for the load-bearing negatives) a second independent implementation in Arb. *Verified*: high-precision numerics with stated grids and precision policies, not a proof. No verified statement is used as a hypothesis of an exact one.

2 The spectral framework

Convention 3 (normalizations). On $(0, \pi)$ define the kernel, entry law, and smoothing operator

$$q_1(u, v) = 2 e^{\beta \cos u \cos v} \sinh(\beta \sin u \sin v) = 4 \sum_{m \geq 1} I_m \sin(mu) \sin(mv), \quad (3)$$

$$s_1(u) = \frac{\beta}{2} \sin u e^{\beta \cos u} = \sum_{m \geq 1} m I_m \sin(mu), \quad (4)$$

$$(Qf)(v) = \frac{1}{2\pi} \int_0^\pi q_1(v, u) f(u) du, \quad (5)$$

so that $Q \sin(m \cdot) = I_m \sin(m \cdot)$. Both closed forms are classical consequences of the generating function $e^{\beta \cos \theta} = I_0 + 2 \sum_{m \geq 1} I_m \cos(m\theta)$; (4) follows by differentiation. All ten verification artifacts in the repository use exactly these normalizations.

Since $q_1 > 0$ on $(0, \pi)^2$ and $s_1 > 0$ on $(0, \pi)$, the walk with step kernel q_1 killed on $\{0, \pi\}$ is well defined; $H(\psi) = \beta \sin(\psi/2) I_1(2\beta \cos(\psi/2))$ and $K_2(\varphi) = I_0(2\beta \cos(\varphi/2))$ appear below. Note $F_B = Q^3 s_1$ and, by the sine orthogonality computation of Lemma 10, F_A is the companion cosine-weighted integral.

3 Theorem A: positivity of F_B

Theorem 4 (exact). $F_B(t) > 0$ for all $t \in (0, \pi)$ and all $\beta > 0$.

First proof (kernel positivity). $F_B = Q^3 s_1$ by (5) and $b_m = m I_m^4$. Each application of Q integrates a positive function against the strictly positive kernel (3); $s_1 > 0$ on $(0, \pi)$. Hence $F_B > 0$ pointwise. $\square$

Second proof (symmetric-decreasing convolution). With $K_2(\varphi) = I_0(2\beta \cos(\varphi/2))$ and $P_4 = K_2 * K_2$ (circle convolution; Section 6), $F_B = -\frac{1}{2}P'_4$. The function K_2 is even and *strictly decreasing* in $|\varphi|$ on $[0, \pi]$ (I_0 strictly increasing, $\cos(\varphi/2)$ strictly decreasing). By the layer-cake representation, $K_2 = \int_0^\infty \mathbf{1}_{A_\lambda} d\lambda$ with A_λ centered arcs whose half-lengths $a(\lambda)$ fill $(0, \pi)$ continuously (strict decrease). For centered arcs, $(\mathbf{1}_A * \mathbf{1}_B)(t) = |A \cap (t+B)|$ is even and nonincreasing in $|t|$ on $[0, \pi]$, and *strictly decreasing* on $[|a-b|, \min(a+b, 2\pi-a-b)]$. Hence $P_4(t) = \iint |A_\lambda \cap (t+A_\mu)| d\lambda d\mu$ is even, nonincreasing on $[0, \pi]$, and strictly decreasing on $(0, \pi)$ because for every $t \in (0, \pi)$ a positive measure of pairs (λ, μ) has $|a(\lambda) - a(\mu)| < t < \min(a+b, 2\pi-a-b)$ (the half-lengths fill $(0, \pi)$). Thus $F_B = -\frac{1}{2}P'_4 > 0$ on $(0, \pi)$. $\square$

4 Theorem B: ratio monotonicity for $0 < \beta \leq 3$

Lemma 5 (exact; Wronskian identity). *For real arrays with $\sum m(|a_m| + |b_m|) < \infty$,*

$$W := 2(F'_A F_B - F_A F'_B) = \sum_{1 \leq m < n} c_{mn} T_{mn}(t), \quad c_{mn} = a_m b_n - a_n b_m, \quad (6)$$

where $T_{mn}(t) = (m-n) \sin((m+n)t) + (m+n) \sin((n-m)t)$.

Proof. Product-to-sum: $T_{mn} = 2[m \cos(mt) \sin(nt) - n \sin(mt) \cos(nt)] =: 2G(m, n)$, antisymmetric with $G(n, n) = 0$; since c_{mn} is antisymmetric, $\sum_{m < n} c_{mn} 2G = 2 \sum_{m, n} a_m b_n G(m, n) = 2[F'_A F_B - F_A F'_B]$ (first recorded in [7]). $\square$

The coefficient input is the weighted Turán-type lemma (Appendix A, self-contained): $\varphi_m = a_m/b_m$ is strictly increasing in $m \geq 1$ for every $\beta > 0$, hence *every minor is negative*: $c_{mn} < 0$ for $m < n$ (exact).

Lemma 6 (exact; the pair identities and bounds). *For $t \in (0, \pi)$, with $p = n - m$, $q = n + m$:*

$$(a) \quad T_{12} = 4 \sin^3 t; \quad T_{13} = 16 \sin^3 t \cos t; \quad T_{24} = 8 \sin^3(2t), \text{ so } |T_{13}| \leq 16 \sin^3 t \text{ and } |T_{24}| \leq 64 \sin^3 t.$$

$$(b) \quad T_{m, m+1} \geq 0 \text{ for every } m \geq 1.$$

$$(c) \quad |T_{mn}| \leq \frac{\pi^3}{48} pq(q^2 + p^2) \sin^3 t \text{ for all } m < n.$$

Proof. (a) $3 \sin u - \sin 3u = 4 \sin^3 u$ gives T_{12} (at $u = t$) and T_{24} (at $u = 2t$); $T_{13} = 4 \sin 2t - 2 \sin 4t = 4 \sin 2t(1 - \cos 2t) = 16 \sin^3 t \cos t$. (b) $T_{m, m+1} = (2m+1) \sin t - \sin((2m+1)t) \geq 0$ since $|\sin(qt)| \leq q \sin t$ on $(0, \pi)$ (induction on q). (c) T_{mn} and its first two derivatives vanish at $t = 0$ and $t = \pi$ (the parities of $m \pm n$ agree), and $|T'''_{mn}| \leq pq^3 + qp^3 = pq(q^2 + p^2)$; hence $|T_{mn}(t)| \leq \frac{pq(q^2 + p^2)}{6} \min(t, \pi - t)^3$, and $\sin t \geq \frac{2}{\pi} \min(t, \pi - t)$ gives the claim. $\square$

Theorem 7 (exact modulo the certified computation `certify_thmB.py`). *For $0 < \beta \leq 3$: $W < 0$ on $(0, \pi)$, hence $E' < 0$ on $(0, \pi)$.*

Proof. Drop the adjacent pairs $m \geq 2$ (each contributes $c_{m, m+1} T_{m, m+1} \leq 0$ by Lemma 6(b) and $c < 0$); keep T_{12} , T_{13} , T_{24} exactly and majorize every other pair by Lemma 6(c). Pointwise in t ,

$$\frac{W(t)}{\sin^3 t} \leq \text{Crit}(\beta) := -4|c_{12}| + 16|c_{13}| + 64|c_{24}| + \frac{\pi^3}{48} \sum_{\substack{p \geq 2 \\ (m, n) \neq (1, 3), (2, 4)}} pq(q^2 + p^2) |c_{mn}|.$$

The certificate `certify_thmB.py` (Section 9) encloses Crit in interval arithmetic over β -boxes covering $[1/20, 3]$ (57 boxes, adaptive; I_m increasing in β gives rigorous boxes; cutoff $M = 60$ with explicit tail) and certifies $\text{Crit} < 0$ on all of $[1/20, 3]$; a stability pass at elevated precision re-certifies the tightest boxes. For $0 < \beta \leq 1/20$, Lemma 8 applies. $\square$

Lemma 8 (exact; small β). *For $0 < \beta \leq 1/20$: $\text{Crit}(\beta) < 0$.*

Proof. Let $L_m = (\beta/2)^m/m!$ and $U_m = e^{\beta^2/4}L_m$, so $L_m \leq I_m \leq U_m$ (positive series). Then $|c_{12}| = a_2b_1 - a_1b_2 \geq L_1^6L_2^2 - 4e^{2\beta^2}L_1^2L_2^6 = \frac{1}{4}(\beta/2)^{10}(1 - e^{2\beta^2}(\beta/2)^4) \geq \frac{1}{4}(\beta/2)^{10}(1 - 10^{-6})$, using $a_2 \geq I_1^2I_2^2$, $b_1 \geq L_1^4$, $a_1 = 2I_1^2I_2^2 \leq 2U_1^2U_2^2$, $b_2 = 2I_2^4 \leq 2U_2^4$. Every non-adjacent pair has $|c_{mn}| \leq a_nb_m + a_mb_n$ with total $(\beta/2)$ -degree at least 14; since $a_m \leq 2mU_{m-1}^2U_m^2$ and $b_m \leq mU_m^4$, the ratio of consecutive weighted terms in either index is at most $(\beta/2)^2 \cdot (\text{polynomial factor}) \leq 1/100$ at $\beta \leq 1/20$, so the full weighted sum is at most twice its leading term and $\frac{\pi^3}{48} \sum pq(q^2+p^2)|c_{mn}| \leq 10^3(\beta/2)^{14}$ (a generous count of the (1, 3), (1, 4), (2, 4) leading terms). Hence $\text{Crit} \leq -(\beta/2)^{10}(1 - 10^{-6} - 4 \cdot 10^3(\beta/2)^4) < 0$ since $(\beta/2)^4 \leq (1/40)^4$. $\square$

Remark 9 (verified; the sharp family and saturation). The sharper per-pair bound $|T_{mn}| \leq \frac{pq(q^2-p^2)}{6} \sin^3 t$ and the adjacent floor $T_{m,m+1} \geq 2m \sin^3 t$ (note the sign: the floor is a *lower* bound on a positive quantity) are verified far beyond the range used ($q \leq 151$; 13 pairs to (10, 30), max ratio 1.0) and would extend Theorem 7 to $\beta \approx 3.5$; their per-pair certification is a scheduled upgrade. Beyond that the decomposition is *saturated*: with the exact far block (cancellation included) the criterion fails at $\beta = 6$ by $+2.2 \times 10^{14}$ while the true $W/\sin^3 t$ remains negative (-6.9×10^{12}) — a verified adversarial finding about the method, and the reason the large- β closure uses different machinery (Sections 6–8).

5 The bridge structure of E (exact)

Let $k = q_1 \circ q_1$ denote the two-step kernel, $k(\psi, t) = 4 \sum_{m \geq 1} I_m^2 \sin(m\psi) \sin(mt)$, and define

$$\omega_t(d\psi) \propto (Qs_1)(\psi) k(\psi, t) d\psi \quad \text{on } (0, \pi), \quad (7)$$

the law of the time-2 position (midpoint) of the four-step bridge entering from 0 with law s_1 and conditioned to end at t .

Lemma 10 (exact). $\int_0^\pi (Qs_1) k(\cdot, t) d\psi = 2\pi F_B(t)$ and $\int_0^\pi \cos \psi (Qs_1) k(\cdot, t) d\psi = \pi F_A(t)$.

Hence, as an identity,

$$E(t) = \mathbb{E}_{\omega_t}[\cos \psi] = \frac{F_A(t)}{2F_B(t)}.$$

Proof. $(Qs_1)(\psi) = \sum m I_m^2 \sin(m\psi)$. By sine orthogonality the first integral picks the diagonal, giving $2\pi \sum m I_m^4 \sin(mt)$. In the second, $\cos \psi \sin(m\psi) \sin(n\psi)$ integrates to $\frac{\pi}{4}(\delta_{n,m+1} + \delta_{n,m-1})$, which produces exactly the weights $(n-1)I_{n-1}^2 + (n+1)I_{n+1}^2$ of a_n . $\square$

Lemma 11 (exact; entry law). $\partial_t k(0^+, \psi) = 2H(\psi)$, where $H(\psi) = \beta \sin(\psi/2) I_1(2\beta \cos(\psi/2))$.

Proof. Differentiate in ψ the squared Graf identity $I_0^2 + 2 \sum_{m \geq 1} I_m^2 \cos(m\psi) = I_0(2\beta \cos(\psi/2))$ [4, §10.23(ii)] and compare with $\partial_t k(0^+, \psi) = 4 \sum m I_m^2 \sin(m\psi)$. $\square$

Proposition 12 (certified negative; no monotone full-path coupling). *At $\beta = 20$ the time-3 marginal $\rho_t(u) \propto (Q^2 s_1)(u) q_1(u, t)$ is not stochastically monotone in t : with $a = \frac{4802}{10^4}$, $t_1 = \frac{29621}{10^4}$, $t_2 = \frac{30070}{10^4}$, the tail masses satisfy $T(t_2, a) < T(t_1, a)$, with certified enclosure*

$$T(t_2, a) - T(t_1, a) \in [-2.5299150980081786690028107, -2.5299150980081786690028074] \times 10^{-10}.$$

Consequently no monotone coupling of full bridge paths exists at $\beta = 20$. The certificate uses exact termwise tails ($J(m, n, a) = \int_a^\pi \sin mu \sin nu du$ in closed form: no quadrature error), positive-series Bessel enclosures with geometric tails, explicit truncation bounds, nested passes $(M, \text{prec}) = (120, 350) \subset (140, 500)$, and two independent interval implementations (`mpmath.iv`; `Arb` via `python-flint`).

Proposition 13 (certified negatives; the weight–kernel matrix). *At $\beta = 20$, for every entry weight $Q^{k-1} s_1$ ($k = 1, 2, 3, 4$) the marginal one q_1 -step from the moving endpoint fails stochastic monotonicity, with certified witnesses sharing $(t_1, t_2) = (3, 61/20)$: $k = 1$ (two-step bridge midpoint, the minimal counterexample), $a = 9/10$, margin $\approx -1.34 \times 10^{-2}$; $k = 2$, $a = 3/4$, $\approx -1.80 \times 10^{-6}$; $k = 4$, $a = 3/10$, $\approx -4.75 \times 10^{-13}$ ($k = 3$ is Proposition 12). Two smoothing steps repair all tested weights (verified); one never does. The statement is relative to the natural weight family: k_2 possesses its own total-positivity corner ($\beta \gtrsim 8$) reachable by two-atom adversarial weights.*

Lemma 14 (exact; the 3/2 mechanism). $\lim_{u \downarrow 0} \frac{\partial_u \partial_t \log q_1(u, t)}{\beta \sin t u} = 1 + \frac{2\beta}{3} \cos t$. *For $t > \pi/2$ the corner sign change of $\partial_u \partial_t \log q_1$ occurs exactly when $\beta |\cos t| > 3/2$ — the same threshold as the TP_2 failure of q_1 .*

Proof. $\partial_t \log q_1 = -\beta \sin t \cos u + \beta \cos t \sin u \coth(\beta \sin t \sin u)$ (direct differentiation of (3)); expand $\coth z = z^{-1} + z/3 + O(z^3)$ and differentiate in u ; the stated limit is a one-line computation (symbolic residual zero). $\square$

Remark 15 (verified; the mechanism reading). The identification of this local defect as *the* engine of Propositions 12–13 — sine-type entry weights, vanishing linearly at 0 with mass on both sides of the non-monotone region, flip the covariance $\text{Cov}_{\rho_t}(\mathbf{1}_{u \geq a}, \partial_t \log q_1)$, while the flat weight does not — is supported by the full weight–kernel matrix and the β -scan of the violations (onset at $\beta \gtrsim 12$, consistent with the threshold), but is a verified reading, not a theorem.

6 The single-Bessel reduction and the master formula

Lemma 16 (exact; Neumann product formulas). *For $x, y \geq 0$, with $w = \sqrt{x^2 + y^2 + 2xy \cos \alpha}$,*

$$I_0(x)I_0(y) = \frac{1}{2\pi} \int_0^{2\pi} I_0(w) d\alpha, \quad I_0(x)I_1(y) = \frac{1}{2\pi} \int_0^{2\pi} I_1(w) \frac{y + x \cos \alpha}{w} d\alpha.$$

Proof. The first is the integrated Graf/Neumann addition theorem [4, §10.23(ii)]; the second is its ∂_y -derivative. $\square$

With $K_2(\varphi) = I_0(2\beta \cos(\varphi/2)) = \sum_{m \in \mathbb{Z}} I_m^2 e^{im\varphi}$ and $P_4 = K_2 * K_2$ (circle convolution), $P_4 = \sum_m I_m^4 e^{imt}$ and $F_B = -\frac{1}{2} P_4'$. Applying Lemma 16 to both convolution factors:

Lemma 17 (exact; master representation). *Let $s = \varphi - t/2$, $c_1 = \cos(\varphi/2)$, $c_2 = \cos((t - \varphi)/2)$, $P = \sin^2(s/2)$, $Q = \sin^2(\alpha/2)$, and*

$$R^2 = c_1^2 + c_2^2 + 2c_1c_2 \cos \alpha = 4[\cos^2(t/4)(1 - P - Q) + PQ]. \quad (8)$$

Then

$$F_B(t) = \frac{\beta \sin(t/2)}{8\pi^2} \iint_{[-\pi, \pi]^2} \frac{I_1(2\beta R)}{R} [\cos^2(s/2) - \sin^2(\alpha/2)] ds d\alpha, \quad (9)$$

and, under the common positive kernel $K = I_1(2\beta R)/R$,

$$E(t) = \frac{\langle \cos(t/2) \cos 2s + \cos \alpha [\cos(t/2) \cos s - \sin^2 s] \rangle}{\langle \cos s + \cos \alpha \rangle}, \quad (10)$$

where $\langle \cdot \rangle$ denotes the $K ds d\alpha$ -integral over $[-\pi, \pi]^2$. The prefactor collapse behind (9) is the sum-to-product symmetrization $\frac{1}{2}[\sin(\frac{t-\varphi}{2})(c_2 + c_1 \cos \alpha) + (\varphi \leftrightarrow t - \varphi)] = \frac{1}{2} \sin(t/2)(\cos s + \cos \alpha)$, and $\cos s + \cos \alpha = 2[\cos^2(s/2) - \sin^2(\alpha/2)]$. All identities are ring identities (symbolic zeros); (9) and (10) match the quartic series with ratio 1.000000000000 at $t \in \{0.4, 1.1, 2.3, 2.9\}$, $\beta = 4$.

Lemma 18 (exact; geometry of the (P, Q) square, $0 < t < \pi$). *On $[0, 1]^2$: the main saddle is $(0, 0)$ with $R = 2 \cos(t/4)$; the secondary saddle is $(1, 1)$ with $R = 2 \sin(t/4)$ and gap $2(\cos(t/4) - \sin(t/4)) > 0$ degenerating only at $t = \pi$; the zero set $\{R = 0\}$ consists of exactly the two corners $(1, 0)$ and $(0, 1)$, and at both of them both brackets of (9) and (10) vanish, so the singular points of K are switched off (the extension $I_1(z)/z \rightarrow \frac{1}{2}$ makes K continuous with value β). At $t = 0$ the zero set degenerates to the edges $P = 1$, $Q = 1$. The saddle deficit is the exact sum of two squares*

$$4 \cos^2(t/4) - R^2 = 4 \cos^2(t/4) \sin^2(s/2) + 4c_1c_2 \sin^2(\alpha/2), \quad c_1c_2 = \cos^2(t/4) - \sin^2(s/2), \quad (11)$$

with sign region $c_1c_2 \geq 0 \iff |s| \leq \pi - t/2$.

Proof. (8) and (11) are product-to-sum computations (symbolic zeros). $R^2 \geq (|c_1| - |c_2|)^2$ with equality forcing $|c_1| = |c_2|$ (i.e. $s = 0$ or $s = \pm\pi$) and $\cos \alpha = \mp \text{sign}(c_1c_2)$, which lands exactly on $(P, Q) \in \{(0, 1), (1, 0)\}$. Bracket vanishing at both points is direct substitution. The Jacobian of $(s, \alpha) \mapsto (P, Q)$ carries the arcsine weight $ds d\alpha = dP dQ / \sqrt{P(1-P)Q(1-Q)}$ (per quadrant), singular at all four corners; near the main saddle one therefore works in (s, α) , and in (P, Q) only for the global geometry. $\square$

Lemma 19 (exact; saddle cancellations). *Write N, D for numerator and denominator brackets of (10), $C = \cos(t/2)$, $c_0 = \cos(t/4)$. At the saddle $(s, \alpha) = (0, 0)$:*

$$\frac{N}{D} \Big|_{(0,0)} = C, \quad N_{\alpha\alpha} - C D_{\alpha\alpha} = 0, \quad N_{ss} - C D_{ss} = -4C - 2,$$

and for $N_t = -\frac{1}{2} \sin(t/2)[\cos 2s + \cos \alpha \cos s]$:

$$N_{t,\alpha\alpha} - m D_{\alpha\alpha} = 0 \quad (m = -\frac{1}{2} \sin(t/2)), \quad N_{t,ss} - m D_{ss} = 2 \sin(t/2).$$

Also exact: $2C + 1 = 4c_0^2 - 1$ and $C c(t) = 2c_0 - 1/(2c_0)$ for $c(t) = (4c_0^2 - 1)/(2c_0 C)$.

Proposition 20 (verified; the second-order laws). *The frozen-kernel expansion built on Lemma 19 (with $\langle s^2 \rangle = 1/(\beta c_0) + O(\beta^{-2})$ from (11)) predicts*

$$E = C \left(1 - \frac{c(t)}{\beta}\right) + O(\beta^{-2}), \quad E' = -\frac{1}{2} \sin(t/2) + \frac{\sin(t/4)(4c_0^2 + 1)}{8\beta c_0^2} + O(\beta^{-2}). \quad (12)$$

The coefficient identities are exact (Lemma 19); the laws with remainder are verified, not yet certified: measured remainders are $\approx 0.15/\beta^2$ with true convergence at fixed t (e.g. β^2 -scaled residues 0.1776/0.1767/0.1762 at $t = 1, \beta = 30/60/120$), and $\beta(E' + \frac{1}{2} \sin(t/2)) \rightarrow \sin(t/4)(4c_0^2 + 1)/(8c_0^2)$ at $t = 1, 2$. The remainder control is exactly what the closure map owes. Note the $1/\beta$ correction to E' is positive: it opposes the target inequality and must be beaten, not used. The α -cancellations imply every kernel correction multiplies a vanishing fluctuation and enters at $O(\beta^{-2})$ only.

Remark 21 (exact decomposition of E'). With the normalized positive measure $d\mu \propto K ds d\alpha$, $E' = \langle N_t \rangle / \langle D \rangle + \langle (N - ED) \partial_t \log K \rangle / \langle D \rangle$, $D_t = 0$, and the regular form $\partial_t \log K = 4cc'(1 - P - Q)[zI_0(z)/I_1(z) - 2]/R^2$, $z = 2\beta R$, extends continuously by $4\beta^2 cc'(1 - P - Q)$ at $R = 0$. Verified split at $(\beta, t) = (5, 1.3)$: $-0.238212 - 0.024552 = -0.262764 =$ the finite difference exactly. Bounds for E' are taken from this exact quotient — never by differentiating an estimate of E .

7 The π -endpoint package

Lemma 22 (exact; telescoping). *With a_m as in (1):*

$$A_\infty := \sum_{m \geq 1} (-1)^{m+1} m a_m = 0, \quad \sum_{m \geq 1} (-1)^{m+1} m^3 a_m = - \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2.$$

Hence $F_A(\pi - \varepsilon) = c_3 \varepsilon^3 + O(\varepsilon^5)$ with

$$c_3 = \frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2. \quad (13)$$

Proof. Split $m^r a_m = m^r(m-1)I_{m-1}^2 I_m^2 + m^r(m+1)I_m^2 I_{m+1}^2$ and shift $m \rightarrow m+1$ in the first sum: for $r = 1$ the two weights cancel ($m(m+1) - m(m+1) = 0$); for $r = 3$ they combine to $m(m+1)[m^2 - (m+1)^2] = -m(m+1)(2m+1)$. Absolute convergence justifies the reindexing; $\sin(m(\pi - \varepsilon)) = (-1)^{m+1} \sin(m\varepsilon)$ expands the sine. $\square$

Lemma 23 (exact; integral form and parity). *Let $g(u) = \frac{1}{2} I_1(2\beta \cos u) = \sum_{m \geq 0} I_m I_{m+1} \cos((2m+1)u)$ (addition theorem). Then*

$$c_3 = \frac{S_3 - S_1}{24}, \quad S_1 = \frac{2}{\pi} \int_0^\pi g\left(\frac{\pi}{2} - u\right) g'(u) du, \quad S_3 = \frac{2}{\pi} \int_0^\pi g'\left(\frac{\pi}{2} - u\right) (-g''(u)) du.$$

Moreover $g(-u) = g(u)$ and $g(\pi - u) = -g(u)$ (parity of I_1), hence $g'(\pi - u) = g'(u)$, $g''(\pi - u) = -g''(u)$, and both integrands are invariant under $u \mapsto \pi - u$: the two halves $[0, \pi/2]$ and $[\pi/2, \pi]$ contribute exactly equally, and one may integrate a half range and double. In particular no cancellation between the two phase maxima $u = \pi/4$ and $u = 3\pi/4$ is possible — exactly, not asymptotically.

Proposition 24 (verified; prefactor law). $c_3 > 0$ for $\beta \in \{1, 5, 20, 40, 80, 120\}$ at cancellation-adjusted precision (the alternating sum cancels at scale $e^{-(4-2\sqrt{2})\beta}$; evaluations use $\text{dps} \geq 0.6\beta + 50$), and

$$c_3 \sim A\beta^{3/2}e^{2\sqrt{2}\beta}, \quad A = \frac{1}{24 \cdot 2^{1/4}\pi^{3/2}} = 0.0062922\dots,$$

with measured ratios 0.0058646, 0.0060049, 0.0060996, 0.0061475 at $\beta = 40, 60, 90, 120$ and $O(1/\beta)$ gaps. The certified large- β branch (pending) must produce the signed prefactor with an explicit B : $I_{\text{half}} \geq \frac{A}{2}\beta^{3/2}e^{2\sqrt{2}\beta}(1 - B/\beta) - T(\beta)$, $c_3 = 2I_{\text{half}}$, read β^* , and close $[15, B_{\text{fin}}]$, $B_{\text{fin}} = \max\{\beta^*, 10C\}$, by intervals.

8 The closure map and the candidate threshold

Proposition 25 (verified truth map; all statements on the stated finite grids). Let $g(t, \beta) = E' + \frac{1}{4}\sin(t/2)$. At $\beta = 15$, $g < 0$ at every point of a 13-point grid spanning $[0.05, \pi - 0.09]$ (independently reproduced on a much finer mesh at 110 digits), with thinnest margin -0.00573 at $t = 0.05$ (on the seal window $[0.05, \pi - 0.10]$) and -0.00216 at $\pi - 0.09$; the crossing sits at $\pi - t^* = 0.0890502225$. At $\beta = 12$ the truth itself fails at $\pi - 0.10$ ($g = +0.025$); crossings scale as the mirror window: $\pi - t^* = 0.1155, 0.0891, 0.0644$ at $\beta = 12, 15, 20$, matching $\sqrt{2}/\beta$. A sweep $\beta = 15 \dots 60$ keeps the maximum at $t = 0.05$, drifting to -0.0061 : strong uniformity evidence, not proof.

Conjecture 26 (quantified closure; the seal). Part (ii) of Conjecture 1 holds with the two-scale decomposition: (a) $0 < \beta \leq 3$ by Theorem 7; (b) $\beta \in [3, 15]$ by certified evaluation (the holonomic structure of Section 9 makes the compact cheap); (c) $\beta \geq 15$ by three uniform blocks: left edge $(0, 0.05]$ ($e_2 > 0$, proved by the Chebyshev-sum corollary of the Turán lemma, plus a remainder radius 0.05); bulk $[0.05, \pi - 0.10]$ (enclosure of $\langle s^2 \rangle$ and of the covariance of Remark 21, with the five prototype-to-certificate conversions); right edge with moving boundary: assembly to $\pi - C/\beta$, then the quadratic regime of c_3 on $[\pi - C/\beta, \pi]$ with $C_{\text{mirror}} \leq C \leq C_{\text{quad}}$ (a fixed quadratic window is provably impossible: $\kappa \approx 0.11\beta$ grows while $E(\pi - 0.10)$ stays bounded; retention $E/\kappa d^2$ at $d = 2/\beta$ is 0.896, 0.892, 0.890 at $\beta = 20, 40, 80$ — the validity radius scales as $1/\beta$). The named constants still owed: B , β^* , C_{quad} , C_{mirror} , and the remainder function $T(\beta)$.

The first-order competition is not the obstacle: the two-term law (12) requires only $\beta > (1 + 1/(4c_0^2))/c_0 \leq 3\sqrt{2}/2 \approx 2.121$; everything above that is rigor slack. The candidate $\beta_0 = 15$ is near truth-optimal for its edge window (Proposition 25), not padded.

9 Verification, certificates, and reproducibility

All numerical artifacts live in the repository

<https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>

(directories `scripts/`, `docs/`); the immutable commit hash for the version audited with this manuscript is recorded in the repository's hash registry (`docs/SURFACE-CLOSURE-NOTES.md`) and in the arXiv comment field on submission. Each certificate embeds its pinned statement and self-verifies on every execution.

Certificates (interval arithmetic, outward rounding, nested or stability passes). `certify_time3_negative.py` and its Arb twin (Proposition 12); `certify_bridge_matrix.py` and twin (Proposition 13); `certify_thmB.py` (Theorem 7: 57 adaptive β -boxes covering $[1/20, 3]$, single implementation with an elevated-precision stability pass on the tightest boxes; an Arb twin is a scheduled upgrade). Bessel enclosures use the positive series with an exact-integer stopping rule and geometric tails; floats appear only in loop-termination heuristics. Nested enclosures prevent silent numerical or truncation degradation; statement auditing and the independent Arb implementation guard against common-mode errors.

Verified computations. The master-formula and reduction identities (ratios 1.0 to 12 digits); the truth map of Proposition 25 (dps 80, confirmed independently at 110 digits); the c_3 ladder (Proposition 24); the term-split coefficients of (12) at $\beta = 60, 120, 240$.

Precision policies (each traceable to a caught artifact): $\text{dps} \geq 2.2\beta + 40$ for any evaluation with $t \geq 2$, $\beta > 30$ (mirror cancellation); $\text{dps} \geq 0.6\beta + 50$ for c_3 -type alternating sums; series cutoffs scale with the maximal argument, and scaled-asymptotic evaluation of $e^{-z}I_n(z)$ is the default for $z > 35$; every measured coefficient carries a β -scaling sanity check; whatever can be computed from a positive integral representation is.

Audit trail. Fifteen numerical or formulation artifacts (“ghosts”) were caught during fabrication and writing by the three-way review protocol; Appendix B tabulates them. We consider the table part of the result: it is the reproducibility record that makes the certificates credible.

Machine-checked lemmas. The unit-step and Turán-type coefficient inequalities behind Theorem 7 are formalized in Lean 4/Mathlib (`PhiMonotone.lean`, `FHBesselAmos.lean`, `WronskianIdentity.lean`; axiom oracle [`propext`, `Classical.choice`, `Quot.sound`], no `sorry`). Mathlib has no Bessel functions: what is machine-checked is the ordered-field implication, with the analytic inputs (three-term recurrence [4, 10.29.1]; the Amos-type bound of [1, 2, 3], cited verbatim, shift $\nu \rightarrow \nu + 1$ in an appendix of [6]) carried as named hypotheses. Correct claim everywhere: *machine-checked modulo classical Bessel inputs carried as named hypotheses.*

Acknowledgements and provenance

The conjecture, the bridge reading, the reduction, and all certificates were produced inside an audit-first programme in which three independent reviewing voices verify every claim before acceptance; scores, objections, and the complete ghost ledger are archived with the repository history. Nothing in this paper rests on trust in any single computation: every load-bearing negative has two witnesses, and every identity marked exact has a symbolic-zero transcript.

A The weighted Turán-type lemma, self-contained

Theorem 27 (exact). *For every $x > 0$ and every real $m \geq 1$, with $\varphi_m = a_m/b_m$ built from (1) at argument x : $\varphi_m < \varphi_{m+1}$. Equivalently, with $\rho_\mu = I_{\mu+1}/I_\mu$,*

$$\frac{1}{m} \left(\frac{m-1}{\rho_{m-1}^2} + (m+1)\rho_m^2 \right) < \frac{1}{m+1} \left(\frac{m}{\rho_m^2} + (m+2)\rho_{m+1}^2 \right).$$

Proof. Two classical inputs, valid for all real orders: the three-term recurrence $\rho_{m-1}^{-1} = \rho_m + 2m/x$, $\rho_{m+1} = \rho_m^{-1} - 2(m+1)/x$ [4, 10.29.1], and the Amos-type bound

$$\rho_m(x) < U_m(x) := \frac{x}{a + \sqrt{a^2 + x^2}}, \quad a = m + \frac{1}{2},$$

proved best of its form for all real orders ≥ 0 by Segura [2]; we use the version of [3, Thm. 2] with the shift $\nu \rightarrow \nu + 1$ (the shifted statement is exactly the displayed bound; the two-line shift computation is in [6], App. A). Substitute the recurrences, write $u = \rho_m \in (0, 1)$, $c = 1/x$, $S = u + 1/u$, $P = 1/u - u$; clearing the positive denominators, the claim becomes $\Delta_m > 0$ where the *exact factorization*

$$\frac{\Delta_m}{2m(m+1)} = (S - 3c)(P - (2m+1)c) + (2m+1)c^2$$

holds as an algebraic identity (expand and collect; the coefficients of $u^{\pm 2}$ are $\pm 2m(m+1)$, the linear-in- c terms are $-4m(m+1)c[(m-1)u + (m+2)/u]$, and $(m-1)u + (m+2)/u = \frac{1}{2}[(2m+1)S + 3P]$, $1/u^2 - u^2 = PS$). The second factor is positive because U_m is the positive root of $1/w - w = (2m+1)/x$ (the calibration $1/U_m - U_m = 2a/x$) and $w \mapsto 1/w - w$ is strictly decreasing, so $\rho_m < U_m$ gives $P > (2m+1)c$. The first factor is positive because $\sqrt{a^2 + x^2} > a$ gives $U_m < x/(2a)$, so $u < x/(2m+1) \leq x/3$ and $S > 1/u > (2m+1)/x \geq 3c$. The third term is positive. Strictness is strict throughout. (This is the proof of [6], reproduced so that Theorem 7 is self-contained; the ordered-field implication is machine-checked in `PhiMonotone.lean`, modulo the two classical inputs carried as named hypotheses.) $\square$

B Audit table (the ghost ledger)

Each row: what was wrong, how it was caught, and the rule it produced. All entries are documented with transcripts in `docs/SURFACE-CLOSURE-NOTES.md`.

#	artifact	catch / rule produced
1	random TP_2 scan missed adversarial corners	targeted corner search; adversarial grids mandatory
2	verification cell with unbound parameter	explicit binding rule for inline prints
3	near- π scan at low precision, false violation	precision rule $\text{dps} \geq 2.2\beta + 40$ for $t \geq 2$, $\beta > 30$
4	stale-PDF claim misattributed between voices	provenance ledger: attribute to exact source
5	float64 CDF increments ± 11 (impossible)	positive-representation preference
6	$t=2$, $\beta=240$ residue 0.363 vs 0.317	boundary $t \geq 2$ included in the dps rule
7	I_1 series truncated at 60 terms, $z \leq 480$	cutoffs scale with argument; scaled asymptotics for $z > 35$; β -scaling checks
8	c_3 sign flip at $\beta = 120$ (cancellation)	alternating-sum rule $\text{dps} \geq 0.6\beta + 50$
9	$R = 0$ corner misplaced at $(1, 1)$	complete critical-point census required in prose too
10	one-saddle prefactor off by exactly 2	census rule; the factor 2 later proved an exact parity
11	I_1 lower bound dropped the negative tail	sign audits are a separate mandatory pass
12	fixed quadratic window at π was provably false	interrogate obligations before attempting them
13	Thm. A second proof: rearrangement misquoted	layer-cake arc proof substituted (this revision)
14	Lemma sign error: floor stated as upper bound	corrected; per-pair statements now carry signs explicitly
15	“exact” labels on verified inequalities	tricotomy re-audit of every label (this revision)

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